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SUPERSEDING
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FEDERAL SPECIFICATION

# LADDERS, SAFETY STEP (METAL, ROLLABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers safety metal stepladders mounted on casters. The casters are retractable by a lever arrange or by the application of the weight of a person on the ladder.

1.2 Classification.

1.2.1 Types and classes. Ladders shall be of the following types and classes, as specified (see 6.2):

Type I - Steel.
II - Aluminum.

Class 1 - Wide base.
2 - Narrow base.

1.2.2 Size. Ladders shall be of the following sizes as specified (see 6.2):

Class 1 - 1 to 12 steps inclusive, in one step increments.
Class 2 - 1 to 7 steps inclusive, in one step increments.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple Wall.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-132 - Crate, Wood, Open; Maximum Capacity 2,500 Pounds.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. Steel ladders shall be of plain carbon or alloyed steel at the manufacture's option.

3.1.2 Type II. Aluminum ladders shall be of aluminum alloys 6061 or 6063 of a T4, T5, or T6 temper at the manufacturer's option.

3.1.3 Casters for types I and II ladders. Caster shall be natural (NR) or synthetic (SBR) rubber, or any suitable molded plastic.

3.1.4 Tips for ladder legs. The tips for the ladder legs shall be made from NR, SBR, or any suitable plastic material.

3.2 Ladder frame construction.

3.2.1 Type I ladders. Unless otherwise specified (see 6.2), both classes of ladders of this type shall be either all-welded completely assembled or all-welded partially assembled, at the option of the manufacturer. When partially assembled ladders are furnished they shall consist of no more than five all-welded sub-assemblies, sufficient bolts and self-locking nuts for assembly, and a clear set of instructions. All ladders shall be capable of passing the tests specified in 4.3 through 4.3.4.2.3.

3.2.2 Type II ladders. Unless otherwise specified (see 6.2), both classes of ladders of this type shall be either all-welded or all-riveted completely assembled, or all-welded or all-riveted partially assembled, at the option of the manufacture. When partially assembled ladders are furnished they shall consist of no more than five all-welded or all-riveted sub-assemblies, sufficient bolts and slef-locking nuts for assembly, and a clear set of assembly instructions. All ladders shall be capable of passing the tests specified in 4.3 through 4.3.4.2.3.

3.2.3 Pipe and tube. Steel and aluminum pipe and tube shall be welded, seamless, extruded or drawn. Shape of tubes shall be round, square or any other suitable cross-section at the manufacturer's option. Minimum dimension of pipe and tube shall be that issued by the manufacturers in the assembly of industrial grade ladders.

3.2.4 Diagonals. Diagonals shall be tubular, flat bar stock, or equivalent section for strength. Size of bar stock shall be 1/8 by 1 inch or greater.

3.2.5 Height and depth. The height and depth of ladders when loaded shall be in accordance with table I.



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TABLE I. Height and depth of ladders, inches

| Size | Height to top step | Depth or length at base minimum |
|---|---|---|
| 1 | 10-12 | 13 |
| 2 | 18-20 | 19 |
| 3 | 27-30 | 25 |
| 4 | 26-40 | 30 |
| 5 | 45-50 | 36 |
| 6 | 54-60 | 42 |
| 7 | 65-70 | 47 |
| 8 | 75-80 | 55 |
| 9 | 84-90 | 60 |
| 10 | 94-100 | 66 |
| 11 | 103-110 | 72 |
| 12 | 113-120 | 78 |

3.2.6 Base width. Ladder frames shall have base widths in accordance with table II, as specified (see 6.2):

TABLE II. Base widths, inches

| Class | Size | Base width 1/ |
|---|---|---|
| 1 | 1 to 5 | 25 1/2 through 36 1/2 |
| 1 | 6 to 12 | 30 through 36 1/2 |
| 2 | 1 to 7 | 17 1/2 through 24 |

1/ Outside to outside dimension at bottom of side rails.

3.2.7 Height to base ratio. The maximum height of the top step, when loaded, shall not exceed 4 times the rear base dimension.

3.2.8 Standing area. All standing areas shall be contained within the base dimension.

3.2.9 Hand rails. Hand rail as defined here-in is the upper member of the ladder above the steps and platform. It generally parallels the slope of steps and encompasses the top step or platform on the three exposed sides. When specified (see 6.2), ladders shall have hand rails of pipe or tube in accordance with 3.2.3. Hand rails shall be in accordance with table III, as specified (see 6.2).

TABLE III. Hand rail heights, inches

| Ladder size | Minimum Hand rail height | Maximum Hand rail height |
|---|---|---|
| 2 to 7 | 24 | 30 |
| 8 to 12 | 28 | 42 |

3.3 Steps.

3.3.1 Depth. Steps shall be not less than 7 inches deep except that the top step shall be not less than 8 inches in depth.

3.3.2 Width. Step width shall be approximately equal to the inside distance between side rails.

3.3.3 Spacing. Step rise shall be uniform including distance from ground to first step with spring loaded casters compressed or locking devices actuated. Rise shall be 9 to 10 inches inclusive.

3.3.4 Tread. Step treads shall be expanded metal, corrugated, ribbed, dimpled, embossed, knurled, coated with slip resisting materials, or otherwise treated to minimize the possiblity of slipping.

3.4 Casters. Casters shall have a bolt, rivet or bushing axle and swivel casters shall be furnished with a ball bearing race on the swivel. Unless otherwise specified (see 6.2), casters shall be secured to ladders in a manner which permits their replacement using ordinary tools.

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3.4.1 Caster size. The size of casters shall be in accordance with table IV.

TABLE IV. Minimum caster size, wheel diameter, inches

| Class | Ladder size | Caster size |
|---|---|---|
| 1 | 1-5 | 2 |
| 1 | 6-7 | 3 |
| 1 | 8-12 | 3 or 4 |
| 2 | 1-6 | 1-5/8 or 2 |
| 2 | 7 | 3 |

3.4.2 Caster type, class 2 ladders. Caster for class 2 ladders shall be the swivel type and shall be spring-loaded.

3.4.3 Caster type, class 1 ladders, sizes 1 through 4. Unless otherwise specified (see 6.2), these ladders shall have swivel type, spring-loaded casters.

3.4.4 Caster type, class 1 ladder, sizes 5 through 12. Unless otherwise specified (see 6.2), these ladders shall have two rigid casters in the rear and either one or two swivel type casters in the front.

3.5 Safety locking device. Class 1 ladders in sizes 5 through 12 shall have a positive locking device in the front. This locking device shall be actuated either by pressure on the first step or by the use of a foot-or-hand-operated lever. The front caster or casters when in the disengaged position shall not make contact with the floor surface when swiveled through a circle.

3.6 Ladder legs, all classes. The bottom of all legs shall be tipped with a replaceable rubber or plastic tip which can be replaced with ordinary tools.

3.7 Finish. Steel ladders shall be painted with alumimum or safety yellow paint, as specified (see 6.2). Alumimum ladders may be uncoated, anodized or painted with safety yellow paint, as specified (see 6.2).

3.8 Performance.

3.8.1 Stability. When loaded in accordance with 4.3.2 all legs shall be in contact with the ground or floor and ladders shall remain upright and all legs in contact with the testing surface when the horizontal pull is applied.

3.8.2 Compression strength. Ladders shall not exhibit any permanent deformation or structural failure when tested in accordance with 4.3.3.

3.8.3 Rigidity. The vertical deflection of the ladder when tested in accordance with 4.3.4.2.1 and 4.3.4.2.3 shall not exceed one-quarter inch. The difference between the deflection obtained in accordance with 4.3.4.2.1 and the deflection obtained in accordance with 4.3.4.2.3 shall not exceed one-quarter inch.

3.9 Marking for identification. Each ladder shall be legibly marked by stenciling, stamping, or an affixed name plate with the size and the manufacturer's name, symbol, or trademark of such known charcter that the source of manufacture may be readily determined. The marking will include a statement that the ladder is not to be used around electrical circuits or equipment.

3.10 Workmanship. All exposed metal surfaces shall be free from sharp edges, burrs, or other safety hazards. There shall be no structural defects including deformed steps, handrails, frames, or frame and handrail sections. Rivets in aluminum ladders shall be tightly driven. Welds in steel and aluminum ladders shall be free of cracks and closely spaced in-line surface porosity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling and inspection.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type, class, and size of ladder submitted for inspection at the same time.

4.2.2 Visual, dimensional and workmanship examinations. Ladders shall be examined for these characteristics in accordance with table V and MIL-STD-105 using inspection level S-3 and an Acceptable Quality Level (AQL) of 2.5 per cent defective for major defects and 4.0 cent defective for minor defects.

TABLE V. Classification of defects, end item

| Defects | Major | Minor |
|---|---|---|
| Type, Class, and size not as specified. | X | |
| Inside diameter of tube less than 1/2 inch and pipe less than 1/2 inch IPS. | X | |
| Diagonal bar stock less than 1/8 inch by 1 inch in cross section. | X | |
| Height or base depth and width not in accordance with tables I and II. | X | |
| Handrail height or guardrail height not as required in table III. | X | |
| Step depth less than 7 inches, top less than 8 inches and steps not evenly spaced. | X | |
| Tread on steps not in accordance with 3.3.4. | | X |
| Casters not correct type and size not in accordance with table IV and 3.4.2 through 3.4.4. | | X |
| Class 1, sizes 5-12, positive locking device missing. | X | |
| Painted metal surfaces of ladders not completely covered. | | X |
| Sharp edges or burrs present. | | X |
| Rivets on aluminum ladders not tight. | X | |
| Cracks on in-line surface porosity in welds. | X | |
| Damage or defects affecting function or serviceability. | X | |
| Damage or defects not affecting function or serviceability. | | X |

4.2.3 Sampling for tests.

4.2.3.1 Stability and compression tests. Samples for tests shall be in accordance with level S-2 and AQL of 4 percent defective in accordance with MIL-STD-105.

4.2.3.2 Rigidity test. The sample for the rigidity test will be one size 6 ladder of each design which meets the requirements of this specification. No additional rigidity tests will be required unless a design is changed. The supplier will maintain on file an exact drawing and the supporting test results for the ladders tested in accordance with 4.3.4.1 through 4.3.4.2.3.

4.2.4 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with Table VI. For examination of contents the sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 6 defects per hundred units.

TABLE VI. Classification of preparation for delivery defects

| Examine | Defects |
|---|---|
| Markings (exterior) | Omitted; incorrect, illegible, or improper size, location, sequence, or method of application. |
| Materials | Any component missing, or damage. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container. |

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4.3 Test methods.

4.3.1 Preparation for test. Assemble partially assembled ladders using fasteners that are readily removable, except for the rigidity test.

4.3.2 Stability. Conduct two tests, one with a 90 pound load on the bottom step and the other with the same load on the middle or on the next step higher. Attach a portable scale near the mid-point of the vertical height and pull in a horizontal direction, at an angle of 90 degrees, away from the steps until the scale registers 30 pounds. Sliding or skidding shall not be considered cuase for rejection (see 3.8.1).

4.3.3 Compression. Place a load of 800 pounds at the middle of the top step distributed over an area 6 inches wide by the depth of the step for ladder sizes 1 to 12. Repeat with the same load on the middle or 1 step higher for all ladders size 3 and larger (see 3.8.2).

4.3.4 Rigidity.

4.3.4.1 Assembly of partially assembled ladders. Assemble all partially assembled ladders completely in accordance with the set of instructions specified in 3.2.1 and 3.2.2 using the same type lock fasteners furnished with the assembly. Tighten the fasteners using a uniform amount of torque. The amount of torque required shall be that amount required to properly seat the joint without any visible deformation of the metal.

4.3.4.2 Procedure. Test all fully assembled or partially assembled aluminum and steel ladders in accordance with 4.3.4.2.1 through 4.3.4.2.3.

4.3.4.2.1 Deflection, number 1. Mount the ladder upright on two steel horses placed in the shape of an "L". Clamp both corners of the supported side to the horses. Leave the other side with one corner supported, but unclamped. Leave the remaining corner unsupported. Mark the location of the supported legs on the horses so that the same measuring reference point can be achieved for deflection number 2. Load the unsupported corner with a 50 pound tensile load for one-half minute. (Apply the load to the bottom horizontal siderail as close to the corner as possible.) Release the load and measure the height from the surface of the floor to the bottom of the unsupported corner. Load the unsupported corner with a 250 pound tensile load for a period of one minute. Release the load and repeat the measurement from the surface of the floor to the bottom of the unsupported corner. Subtract one measurement from the other to determine conformance to 3.8.3.

4.3.4.2.2 Vibration. Remove the ladder from the horses and set it upright on a level floor inside a wooden frame. Construct the wooden frame using 2-inch by 4-inch lumber set on edge. The frame shall be large enough to permit a minimum of 1-inch of clearance between the inner surfaces of the frame and the outside edge of the ladder frame. Attach an eccentric-loaded vibrator on one side of the frame at the midway location. Bolt the vibrator to two, one-half inch metal plates straddling the side rail. Use lock washers under each nut. Vibrate the ladder using a range of 40 to 45 Hz inclusive for a period of one hour using a model DV 51/M vibrator, or equal (see 6.3).

4.3.4.2.3 Repeat the procedures of 4.3.2.1 (without tightening any fasteners for the partially assembled ladders) to determine conformance to 3.8.3. The difference between deflection number one, and deflection number two shall not exceed the requirements as specified in 3.8.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A, B or C as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 Cleaning and drying. Surfaces of the casters subject to corrosion shall be cleaned by process C-1 and dried by procedure D-1 of MIL-P-116.

5.1.1.2 Preservation application. Surfaces of the casters and ball bearings that are subject to corrosion shall be coated with type P-7 preservation in accordance with MIL-P-116.

5.1.1.2.1 Wrapping. The coated surfaces of the casters shall be covered or wrapped with material conforming to MIL-B-121, grade A, class 2 and secured with tape conforming to PPP-T-76.

5.1.2 Level C. The ladders shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).



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5.2.1 Level A. Ladders of like description shall be packed in quantities as specified (see 6.2) in close fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant, V3c and PPP-B-640, class 2, grade A. When feasible, the ladders may be packed in a crate conforming to MIL-C-132. Blocking and bracing shall be provided. A shroud will be provided in accordance with the appendix of the crate specification. The fiberboard boxes provided shall be closed in accordance with the appendix applicable box specification. The gross weight of the wood boxes and the triple-wall fiberboard box shall not exceed approximately 200 pounds. The PPP-B-636 fiberboard box shall not exceed the weight limitation of the box specification. Strapping shall be in accordance with the requirements of the applicable box or crate specification.

5.2.2 Level B. Ladders of like description shall be packed in quantities as specified (see 6.2) in close-fitting boxes conforming to PPP-B-585, class 1; PPP-B-501, class 1, PPP-B-601, domestic type; PPP-B-621, class 1; and PPP-B-636, class domestic. When feasible, the ladders may be packed in a crate conforming to MIL-C-132. Blocking and bracing shall be provdied. The fiberboard boxes shall be closed in accordance with the appendix to the applicable box specification. The gross weight of the wood boxes shall not exceed approximately 200 pounds and the fiberboard box shall not exceed the weight limitations of the box specification.

5.2.3 Level C. The ladders shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition markings required by the contractor or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the shipping container shall marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Class 1 ladders. These ladders are intended for industrial use including maintenance operations.

6.1.2 Class 2 ladders. These ladders are intended for interior use in libraries, file rooms, and stock rooms.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, class and size (see 1.2.1 and 1.2.2).
(c) Whether a specific type of construction is required (see 3.2.1 and 3.2.2).
(d) Base width (see 3.2.6).
(e) Whether handrails are required and when specified the height (see 3.2.9).
(f) When casters shall not be replaceable (see 3.4).
(g) When caster types shall be other than as specified (see 3.4.3 and 3.4.4).
(h) Finish (see 3.7).
(i) Size of lot, if different (see 4.2.1).
(j) Selection of applicable level of preservation and packaging required (see 5.1 and 5.2).

6.3 Vibrators. The model DV 51/M vibrator is available from the Martin Engineering Company, U.S. Route 34, Neponset, IL 61345.

DOD coordination has been waived.

Preparing activity:

GSA-FSS

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